AWS Solution Architect Training with
AWS Cloud Practitioner Global Certification

Capstone Projects

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AWS Academy Certified Educator

1. Create an EC2 Windows Instance in AWS

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Objective

EC2 (Elastic Compute Cloud) is a core service provided by Amazon Web Services (AWS) that enables users to rent virtual servers in the cloud to run applications, store data, or handle computational workloads.

An EC2 Windows instance is a virtual machine running the Microsoft Windows operating system on Amazon Web Services (AWS) EC2 (Elastic Compute Cloud). This allows you to run Windows-based applications, manage Windows Server configurations, and use various Microsoft software (like SQL Server, IIS, etc.) in the cloud without having to manage physical hardware.

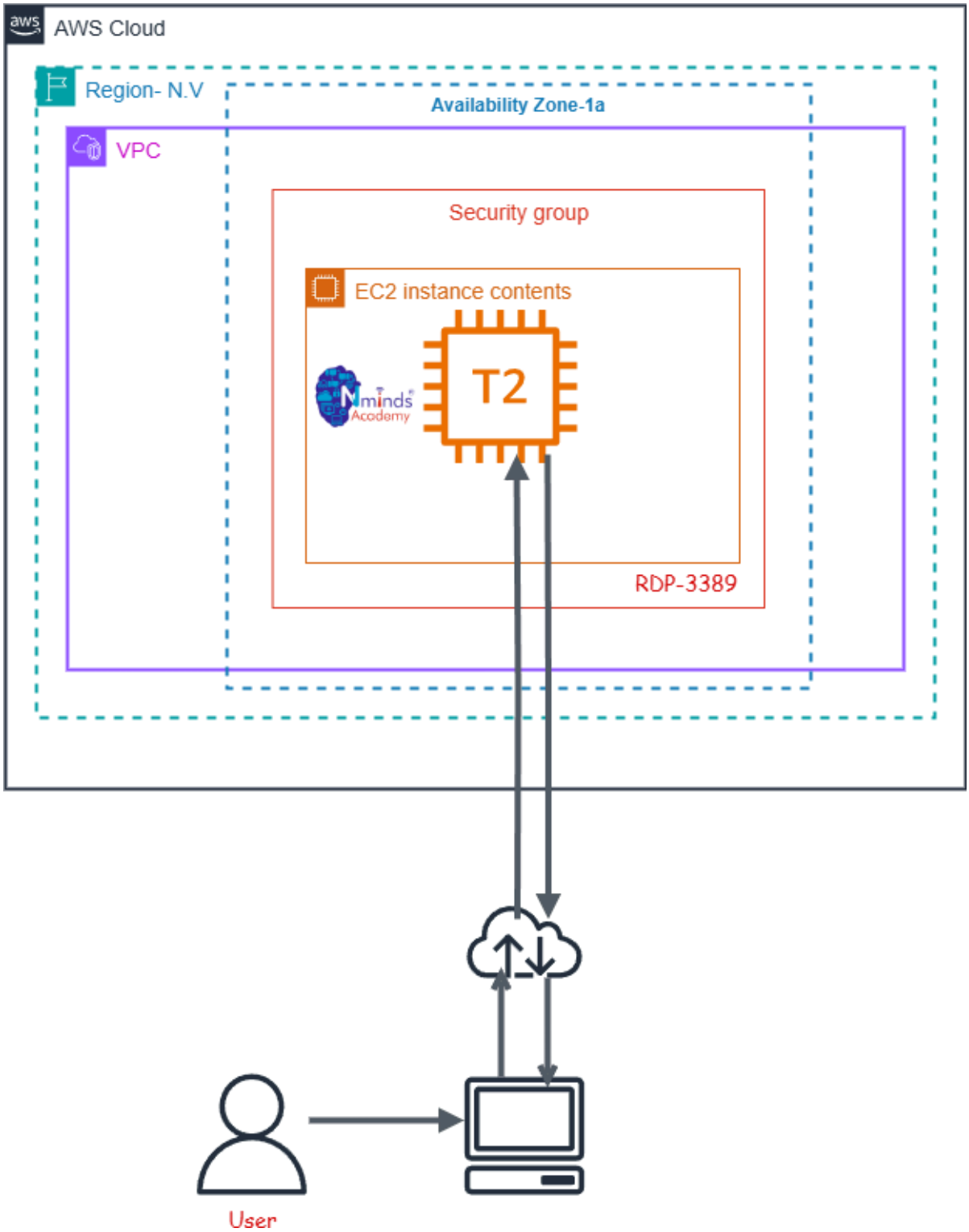
Use Cases for EC2 Windows Instances:

- Web and Application Hosting: Run IIS web servers or other Windows-based web applications in the cloud.
- Database Servers: Host SQL Server databases with EC2 Windows instances, using RDS or EC2 for more control.
- Remote Desktop Applications: Set up RDS (Remote Desktop Services) for remote access to Windows desktops in the cloud.
- Development and Testing: Create development environments with Windows OS to test .NET or other Windows-based applications.

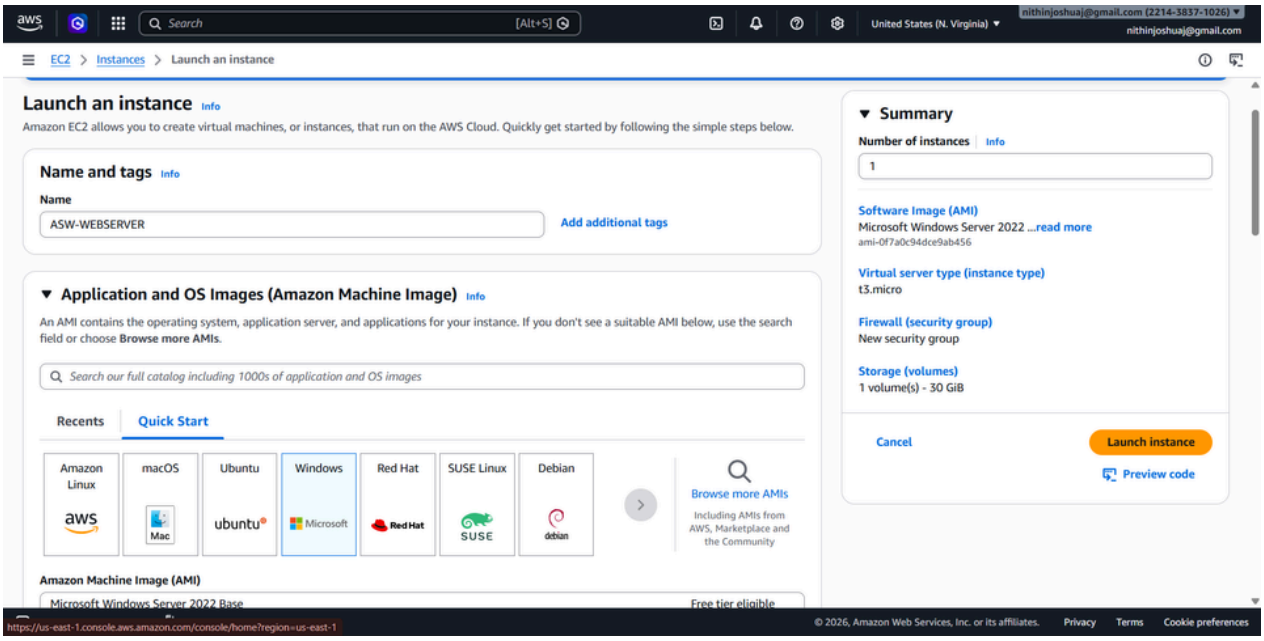
Used Services in AWS:

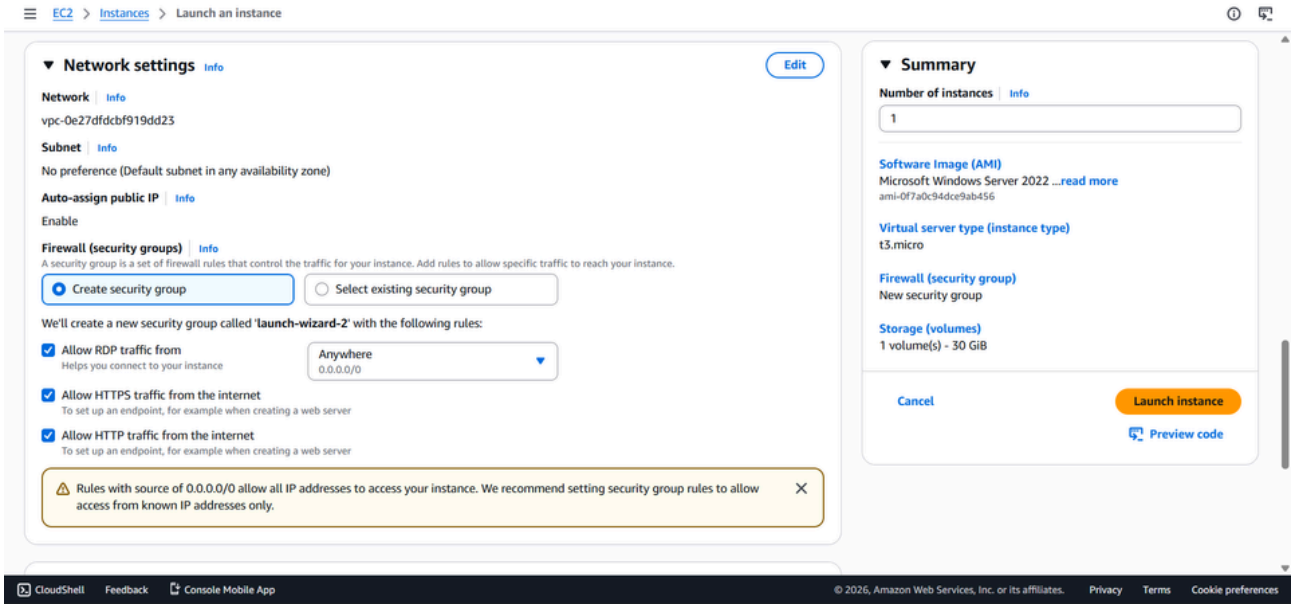
Amazon EC2

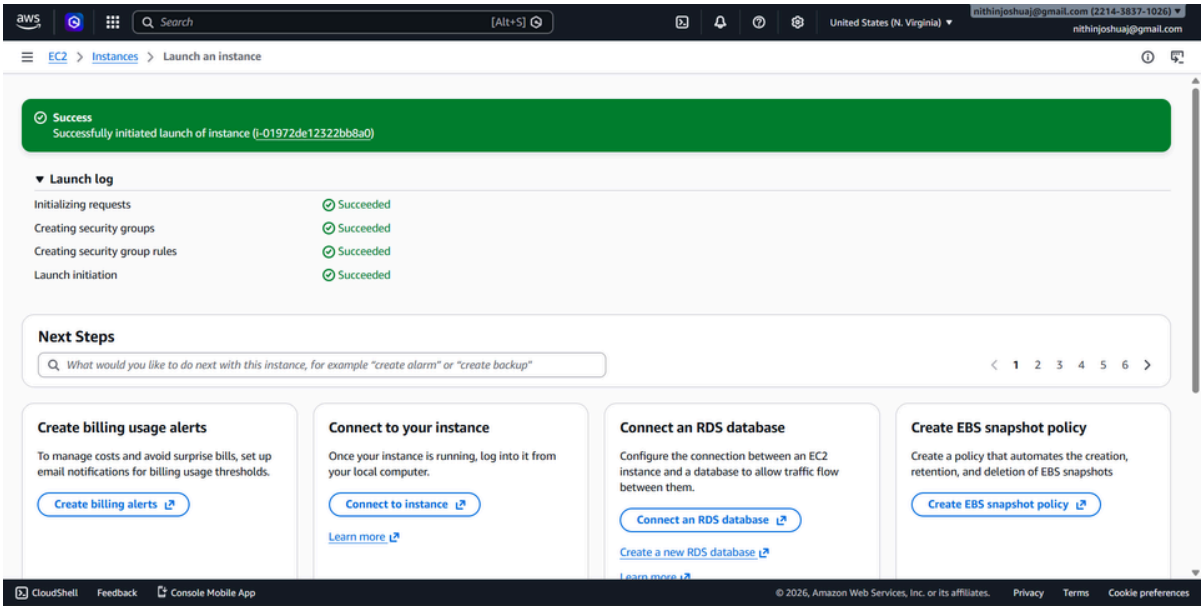
Topology

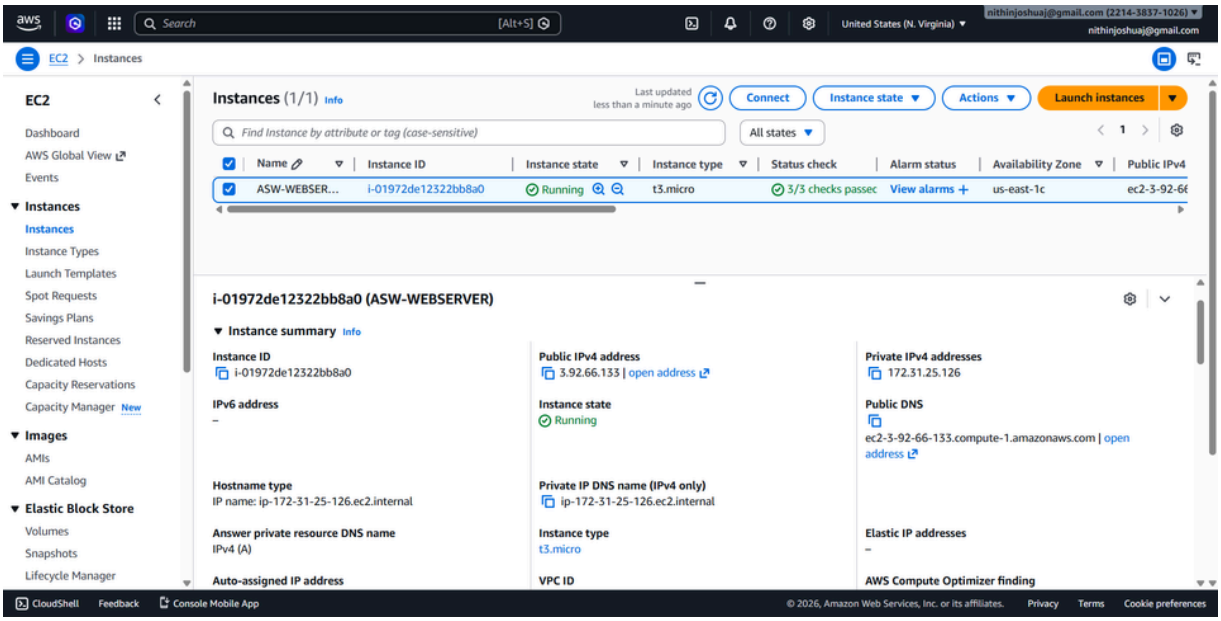

AWS Cloud
Region- N.V
Availability Zone-1a
VPC
Security group
EC2 instance contents
T2
RDP-3389
User

Execution Tasks:






EC2 > Instances > Launch an instance
Success
Successfully initiated launch of instance (i-01972de12322bb8a0)
Launch log
Initializing requests Succeeded
Creating security groups Succeeded
Creating security group rules Succeeded
Launch initiation Succeeded
Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"
Create billing usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.
Create billing alerts
Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more
Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots
Create EBS snapshot policy


EC2 > Instances
Instances (1/1)
Connect
Instance state
Actions
Launch instances
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4
ASW-WEBSER...
i-01972de12322bb8a0
Running
t3.micro
3/3 checks passed
View alarms
us-east-1c
i-01972de12322bb8a0 (ASW-WEBSERVER)
Instance summary
Instance ID
i-01972de12322bb8a0
Public IPv4 address
3.92.66.133 | open address
Private IPv4 addresses
172.31.25.126
IPv6 address
Instance state
Running
Public DNS
ec2-3-92-66-133.compute-1.amazonaws.com | open address
Hostname type
IP name: ip-172-31-25-126.ec2.internal
Private IP DNS name (IPv4 only)
ip-172-31-25-126.ec2.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t3.micro
Elastic IP addresses
Auto-assigned IP address
VPC ID
AWS Compute Optimizer finding